

1985/1. A circle has center on the side AB of the cyclic quadrilateral $ABCD$.

The other three sides are tangent to the circle.

Prove that

$$AD + BC = AB$$

1985/2. Let n and k be given relatively prime natural numbers, $k < n$.

Each number in the set

$$M = \{1, 2, \dots, n-1\}$$

is colored either blue or white.

It is given that

(i) for each $i \in M$, both i and $n-i$ have the same color;

(ii) for each $i \in M$, $i \neq k$, both i and $|i-k|$ have the same color.

Prove that all numbers in M must have the same color.

1985/3. For any polynomial

$$P(x) = a_0 + a_1x + \dots + a_qx^q$$

with integer coefficients, the number of coefficients which are odd is denoted by $w(P)$.

For

$$i = 0, 1, \dots, t,$$

let

$$Q_i(x) = (1+x)^i$$

Prove that if

$$0 \leq i_1 < i_2 < \cdots < i_n$$

then

$$w(Q_{i_1} + Q_{i_2} + \cdots + Q_{i_n}) \geq w(Q_{i_n})$$

1985/4. Given a set M of 1985 distinct positive integers, none of which has a prime divisor greater than 26.

Prove that M contains at least one subset of four distinct elements whose product is the fourth power of an integer.

1985/5. A circle with center O passes through the vertices A and C of triangle ABC and intersects the segments AB and BC again at distinct points K and N , respectively.

The circumscribed circles of the triangles ABC and KBN intersect at exactly two distinct points B and M .

Prove that angle OMB is a right angle.

1985/6. For every real number x_1 , construct the sequence

$$x_1, x_2, \dots$$

by setting

$$x_{n+1} = x_n \left(x_n + \frac{1}{n} \right) \quad \text{for each } n \geq 1$$

Prove that there exists exactly one value of x_1 for which

$$0 < x_n < x_{n+1} < 1$$

for every n .